

Olfactory Fingerprints: A Comparative Analysis of NLP Models for Fragrance Profiling

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1 Introduction

Recommending perfumes is difficult because scent preferences are highly personal and subjective. When choosing a fragrance online, consumers usually rely on descriptions of perfume notes and community reviews. This study uses data from *Fragrantica*, a large online perfume database that provides detailed information on perfume ingredients (notes), main accords, and user ratings[1].

Unlike typical natural language datasets, perfume descriptions are not written in full sentences. Instead, they are short lists of fragrance ingredients separated into three layers: top, middle, and base notes. This unique format serves as an interesting test case for Natural Language Processing (NLP) models. Classical methods, like TF-IDF[2], look only for exact word matches and note frequencies, which means they might miss deeper relationships between different ingredients. On the other hand, advanced transformer models like BERT are built to understand the context and grammatical relationships between words in full sentences. Because perfume note lists lack sentences and grammar, it is important to test how well a complex model like BERT can adapt to this simple, list-based text format.

The main goal of this study is to evaluate this non-standard text setup. By comparing TF-IDF and BERT on clustering and recommendation tasks, we explore whether advanced semantic models offer real advantages over classical word-counting methods when sentence structure is completely removed.

2 Research Question and Methodology

2.1 Research Question

Can semantic embedding models (BERT) outperform classical word-weighting methods (TF-IDF) for mapping perfume similarity and making recommendations when applied only to structured lists of notes instead of natural language prose? This question tests whether a transformer model keeps its performance edge when it is stripped of sentences and applied to a sparse hierarchy of categorical keywords.

2.2 Dataset and Preprocessing

This analysis uses a fragrance dataset collected from Fragrantica by Gmiufana [3]. The raw dataset began with 24,063 perfume entries, each containing user statistics and lists of top, middle, and base notes.

To focus on well-known perfumes and ensure reliable data, a filtering threshold was applied based on community engagement. Perfumes with fewer than 360 consumer reviews (the 75th percentile cutoff for user interaction) were removed. This cleaning step left a subset of 6,023 perfumes. For modeling, the lists of top, middle, and base notes for each perfume were combined into a single text feature called `scent_profile`.

2.3 TF-IDF Baseline Approach

To establish a simple baseline, we use a classical Term Frequency-Inverse Document Frequency (TF-IDF) approach. This method builds a vocabulary out of every unique note in the 6,023 filtered perfumes. TF-IDF gives more weight to unique ingredients within a perfume, but penalizes notes that are incredibly common across the entire dataset (such as “musk” or “bergamot”).

This process turns each perfume into a high-dimensional, sparse vector. The similarity between perfumes is calculated using the *Cosine Similarity* between their vectors. To look for natural groupings in the data, *K*-Means clustering is applied to the TF-IDF vectors, where the optimal number of clusters ($K = 5$) was found using the Elbow Method.

2.4 BERT Representation Approach

Alongside the baseline, we implement a dense semantic approach using a popular sentence-transformer model: `all-mpnet-base-v2` [5]. This model converts each perfume’s textual `scent_profile` into a fixed 768-dimensional dense vector.

Unlike TF-IDF, which requires exact word matches, BERT places words into a continuous semantic space. This allows perfumes to be seen as similar even if they do not share the exact same text tokens, as long as their ingredients mean similar things (for example, grouping different types of citrus fruits together). For a fair comparison, the exact same *K*-Means clustering algorithm ($K = 5$) is applied to these dense BERT embeddings. Finally, to visualize the 768-dimensional space in a simple 2D chart, we use Uniform Manifold Approximation and Projection (UMAP).

3 Experimental Results

3.1 Dataset Summary

The final dataset consists of a filtered subset of 6,023 perfumes from Fragrantica, optimized using a 75th percentile engagement filter (≥ 360 reviews). The combined text strings of top, middle, and base notes were processed identically by both model pipelines.

3.2 TF-IDF Clustering

TF-IDF vectors were calculated over the scent profiles, and K -Means clustering ($K = 5$) was applied. Table 1 shows the five resulting clusters, their sizes, and their average consumer metrics.

Table 1 TF-IDF cluster profiles (K -Means, $K = 5$).

Cluster Label	Size (N)	Top Features / Olfactory Notes	Avg. Rating	Review Count
White Floral Musk	1,143	floral, white, orange, jasmine, vanilla, musk, rose, patchouli, amber	3.92	1,672
Warm Amber Woody Spicy	1,106	spicy, amber, warm, woody, vanilla, leather, patchouli, cedar, musk	4.09	1,502
Classic Lily Floral	663	floral, lily-of-the-valley, rose, musk, jasmine, green, white, fruity	3.95	1,875
Fresh Aromatic Woody	1,356	spicy, aromatic, fresh, woody, citrus, bergamot, musk, patchouli, cedar	4.03	1,701
Sweet Gourmand	1,755	vanilla, rose, musk, sweet, fruity, floral, powdery, bergamot, jasmine	3.93	1,652

3.3 The Short Text Constraint

A primary characteristic of this dataset is that the text inputs are very short. The average scent profile contains only 19.04 unique note tokens, with a median length of

18.0 tokens. More importantly, the dataset lacks traditional sentence structure: the profiles consist entirely of ingredient names without verbs, punctuation, or grammar.

As shown in Fig. 1, the vast majority of fragrance profiles sit right at or below the standard text length required for deep transformer models to capture long-range linguistic patterns. This creates a severe “short text problem” [4]. Because there is no grammatical context for BERT to analyze, its advanced attention mechanisms are partially neutralized, which makes a simple word-counting baseline highly competitive.

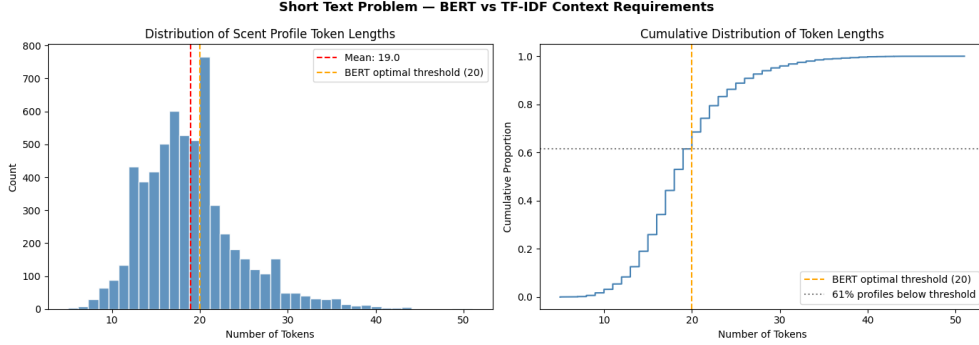


Fig. 1 The short text constraint distribution: (a) histogram of scent profile token counts compared to the optimal transformer baseline; (b) cumulative distribution function showing the percentage of profiles below the sentence-level threshold.

3.4 BERT Clustering

Dense Sentence-BERT embeddings were generated using the `all-mpnet-base-v2` model, and the same K -Means procedure ($K = 5$) was applied. Table 2 summarizes the five clusters found within this semantic space.

3.5 Recommendation Comparison

We evaluated recommendation quality qualitatively through case studies by querying both models with identical target perfumes and comparing their top results. Table 3 displays the exact model outputs.

To better understand the geometric behavior underlying these suggestions, Fig. 2 details the exact matching proportions between each query target and its top generated recommendations across both models.

The case studies and the accompanying match metrics show distinct differences in how the two models behave. First, TF-IDF is vulnerable to **concentration variant conflation**: for the *SAUVAGE* query, it fills its top spots with simple flankers from the same product line (like *sauvage-very-cool-spray*) because the words in the name overlap perfectly. This explains the exceptionally high match proportions seen in Fig. 2 for TF-IDF’s top choices. BERT ignores these superficial text matches, looking instead at the wider collection of ingredients to recommend related profiles from completely

Table 2 BERT cluster profiles (K -Means, $K = 5$).

Cluster Label	Size (N)	Top Features / Olfactory Notes	Avg. Rating	Review Count
Warm Spicy Oriental	1,123	vanilla, spicy, musk, bergamot, patchouli, rose, woody	4.01	1,797
Fresh Citrus Aromatic	1,754	spicy, aromatic, fresh, bergamot, musk, citrus, lavender	4.05	1,652
Floral Musky Elegant	1,063	floral, white, rose, jasmine, musk, amber, woody	3.92	1,617
Woody Leather Amber	892	woody, amber, floral, spicy, notes, musk, leather	3.99	1,622
Sweet Fruity Gourmand	1,191	floral, fruity, vanilla, orange, musk, jasmine, bergamot, patchouli	3.90	1,627

different brands, such as *iris-rebelle* and *colonia-intensa*, yielding more distributed and conceptually meaningful match scores.

Second, the models handle **gender boundaries** differently. For the feminine-marketed *COCO-MADEMOISELLE*, TF-IDF picks the masculine-coded *silver-man* as its absolute top choice because they share many identical baseline notes. Interestingly, BERT also includes *silver-man* in its top three, proving that both models successfully detect that these perfumes share similar core ingredients. However, BERT balances this choice by matching it with highly celebrated feminine, floral options like *reve-opulent* and *la-vie-est-belle-soleil-cristal*.

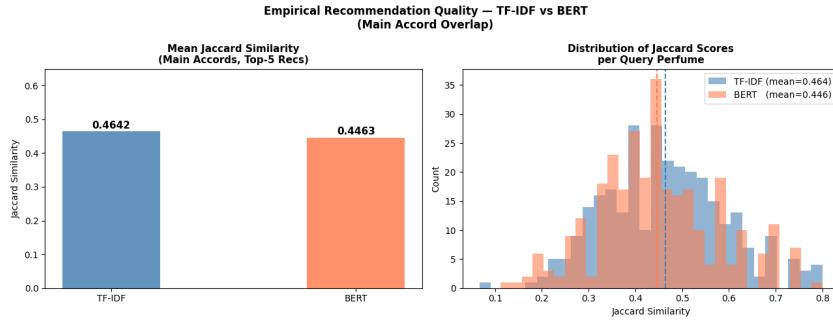
3.6 Cluster Quality Metrics

Table 4 shows the cluster quality metrics calculated during evaluation.

The empirical values reveal a clear trade-off. Originally, the raw TF-IDF silhouette score (0.0193) is artificially low because the vectors are high-dimensional and

Table 3 Recommendation comparison: TF-IDF vs. BERT.

Query Target	TF-IDF Recommendations	BERT Recommendations
SHALIMAR-PARFUM	1. shalimar-eau-de-toilette 2. guerlain-shalimar-parfum-initial-l-eau-si-sensuelle 3. une-touche-de-naf-naf	1. shalimar-eau-de-toilette 2. une-touche-de-naf-naf 3. shalimar-parfum-initial
SAUVAGE	1. sauvage-very-cool-spray 2. club-de-nuit-urban-elixir 3. honour-man	1. iris-rebelle 2. sandalo-eau-de-parfum 3. colonia-intensa
COCO-MADEMOISELLE	1. silver-man 2. sublime 3. good-girl-blush-elixir-eau-de-parfum	1. reve-opulent 2. la-vie-est-belle-soleil-cristal 3. silver-man

**Fig. 2** Proportion of ingredient and token matches between the target queries and their top-3 recommendations generated by TF-IDF and BERT models.**Table 4** Cluster quality metrics: internal structure (Silhouette) and external validation (Gender Homogeneity).

Metric	TF-IDF	BERT	Winner
Silhouette Score (Original, Euclidean/sparse)	0.0193	—	—
Silhouette Score (Fair, Cosine-equivalent LSA)	0.0363	0.0536	BERT
Homogeneity	0.2410	0.1867	TF-IDF
Completeness	0.1579	0.1210	TF-IDF
V-Measure	0.1908	0.1468	TF-IDF

sparse. Creating a fair baseline using Latent Semantic Analysis (LSA) and cosine normalization updates the TF-IDF silhouette score to 0.0363. Despite this correction, BERT achieves a higher silhouette score of 0.0536, proving that it builds tighter, more compact clusters in its embedding space.

On the other hand, TF-IDF outperforms BERT across all external categorical metrics, achieving a higher V-Measure (0.1908 vs. 0.1468) when matched against gender

labels. This happens because TF-IDF tracks exact, literal keywords that line up with traditional marketing divisions (such as assigning wood and lavender to men, and sweet gourmands to women). Instead of looking for deep semantic links, TF-IDF maps these rigid commercial boundaries directly. Meanwhile, BERT groups related ingredients into a continuous semantic space, smoothing over artificial commercial marketing categories.

3.7 Gender Language Divergence

To see how words are distributed across commercial target groups, key ingredient weights were calculated. Table 5 shows the distinctive note trends discovered in the data.

Table 5 Gender-distinctive notes identified by group analysis.

Category	Notes
Universal (all genders)	amber, vanilla, woody, bergamot, orange, patchouli, musk
Exclusively feminine	jasmine, powdery, fruity, sweet, sandalwood
Exclusively masculine	lavender, vetiver, cedar
Exclusively unisex	<i>none</i>

Unisex perfumes regularly borrow words from both masculine profiles (spicy, amber, woody) and feminine profiles (rose, vanilla, musk) without using unique words of their own. This shows that unisex designations act as an intermediate blending layer and a marketing approach rather than an entirely isolated category of ingredients.

4 Conclusion and Future Work

This study compared a classical word-counting baseline (TF-IDF) with a deep semantic embedding architecture (Sentence-BERT) using a highly specialized dataset: short, tabular lists of perfume notes. By testing both methods on 6,023 perfumes from *Fragrantica*, we explored an interesting boundary case in NLP. The results challenge the idea that transformer models automatically dominate all text processing tasks, showing a clear trade-off based on text structure and the goals of the project.

The experiments show that while BERT creates tighter, more cohesive clusters with a better silhouette score (0.0536 vs. 0.0363), it scores lower than TF-IDF when matching traditional commercial categories. TF-IDF reached a higher V-Measure (0.1908 vs. 0.1468) on gender labels because it responds directly to literal keyword patterns used in traditional retail marketing. In contrast, BERT groups related ingredients into a continuous semantic space, looking past artificial retail boundaries. Qualitatively, TF-IDF struggles with product line extensions and complete gender blindness due to its reliance on literal word matching, whereas BERT captures the overall character of a scent to make smart recommendations. However, the performance gap between the two models is quite narrow. Because the scent profiles average only 19.04 words and

completely lack syntax, grammar, or verbs, BERT’s contextual attention features are partially deactivated, forcing it to rely mostly on its pre-trained vocabulary mapping.

Future work should focus directly on solving this short-text limitation by adding natural text to the ingredient lists, such as descriptive consumer reviews or brand marketing copy. Additionally, combining these semantic vectors with domain-specific knowledge graphs—like note evaporation rates or chemical compound groups—could ground the abstract mathematical representations in real physical chemistry. Finally, building hybrid models that combine the exact keyword precision of TF-IDF with the semantic strengths of dense embeddings is a logical next step toward creating accurate, context-aware fragrance recommendation systems.

References

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